Master in Business Analytics & Data Science
Research Capstone in Operations Research

Does Spatial Correlation of Carbon Intensity Improve Distributionally Robust Carbon-Aware Data-Center Scheduling?

A Multi-Grid Falsification Study with a Causal Mechanism

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AI Use Declaration. Generative AI tools (Anthropic Claude) were used as a research and engineering assistant under the author's direction: for code implementation support, experiment orchestration, literature triage, and drafting assistance. All research questions, modelling decisions, experimental designs, and interpretations are the author's, developed with the supervisor; all results were produced by version-controlled, pre-registered code reviewed by the author, and all text was reviewed and edited by the author.

Abstract

Carbon-aware schedulers shift data-center compute toward cleaner hours and regions, and recent work models carbon intensity as uncertain within a distributionally robust optimization (DRO) framework. A natural extension treats carbon intensity as a stochastic *vector* across coupled regions, so that *spatial correlation* can inform the schedule. This thesis asks whether that spatial structure carries any robust scheduling value. Using a Mahalanobis–Wasserstein DRO solved as a second-order cone program, we design a falsification test that compares the schedule fitted to the joint covariance against one fitted to a block-diagonal covariance with cross-region structure destroyed, scored by out-of-sample conditional value-at-risk ($\mathrm{CVaR}_{0.95}$) of daily emissions under pre-registration. Across three US/Canada grids spanning the dependence spectrum (the US West, an Ontario-anchored Eastern Interconnection belt, and an engineered solar/wind/hydro portfolio), with an Iberia–France low-correlation anchor, the answer is a replicated **null**: the spatial gap never exceeds a few tenths of one percent and survives shrinkage, residualization, Benjamini–Hochberg correction, walk-forward validation, and a tighter-ramp sensitivity. Two diagnostics give the mechanism: a *mean-ablation* shows the covariance signal is genuine but dominated by the mean field, and a *tail-dependence* analysis shows the residual dependence is non-elliptical, radially asymmetric and upper-tail independent, hence invisible to covariance-based ambiguity sets. A second phase forecloses the "wrong object" objection: Gaussian, lower-tail Clayton, and the maximal (comonotone) copula all leave the null intact, and an a-priori bound caps the achievable gain. The recommendation is that per-region marginal schedulers capture essentially all the value; spatial value, if any, must come from an active transfer channel, not a richer dependence model.

Contents

1 Introduction

1.1 Context

Data centers consumed an estimated 1–2% of global electricity and their demand is growing rapidly with AI workloads [7]. Unlike most industrial loads, compute is unusually *flexible*: many batch and training workloads can be delayed by hours or moved between sites with little loss of utility. This movement is *logical*, jobs relocate over the network between an operator's data centers, and does not require the regions' power grids to be physically interconnected; spatial correlation of carbon intensity therefore matters as information for the scheduler, not as an electrical constraint. Because the carbon intensity of grid electricity varies strongly by hour and by region, driven by wind, solar, hydro availability, and demand, this flexibility lets a data-center operator act as a grid-aware participant, deferring work from dirty hours to clean ones [17, 20]. Google's carbon-intelligent computing platform demonstrated such temporal shifting at production scale [17], and recent systems work extends it to variable-capacity operation [21].

Scheduling against *tomorrow's* carbon intensity, however, means scheduling under uncertainty. Distributionally robust optimization (DRO) addresses this by optimizing against the worst distribution in an ambiguity ball around the empirical distribution [5], and has been applied to carbon-aware scheduling with carbon intensity as the uncertain quantity [6]. Existing formulations treat each region's carbon intensity essentially in isolation. Yet carbon intensities of electrically interconnected regions co-move: weather fronts traverse neighbouring balancing areas, and shared solar fleets create synchronized clean periods. It is therefore natural to model carbon intensity as a stochastic *vector* with a full spatial covariance, the extension studied here.

1.2 Problem statement and research questions

Modelling spatial dependence is not free: cross-region covariance blocks must be estimated from limited data, and a richer uncertainty model is only worthwhile if it buys measurably better schedules. The literature has assumed, rather than tested, that spatial correlation in the carbon field is worth modelling. This thesis tests that assumption directly.

RQ1. Is the carbon intensity of electrically coupled regions genuinely spatially correlated, beyond shared diurnal cycles?

RQ2. Does feeding that spatial correlation to a covariance-based (Mahalanobis–Wasserstein) DRO scheduler reduce out-of-sample tail emissions ($\mathrm{CVaR}_{0.95}$), relative to an otherwise identical scheduler that ignores cross-region dependence?

RQ3. If not, *why* not, and what dependence structure, if any, would a richer uncertainty model need to capture?

1.3 Contributions

The thesis makes four contributions. (i) *A falsification design*: a shuffled-marginals test that isolates exactly the value of spatial covariance, evaluated out-of-sample under pre-registration discipline (commit-locked configurations, dry-run gates, single test-set read). (ii) *A replicated empirical null across the dependence spectrum*: three primary US/Canada region sets, two interconnected grids where strong correlation survives de-seasonalization (the US West (California, Nevada, Phoenix; and an Ontario–Eastern belt) and one engineered low, heterogeneous-correlation solar/wind/hydro portfolio) corroborated by an Iberia–France panel where residual correlation collapses to the diurnal cycle, all yielding spatial gaps below 0.4% of CVaR, robust to estimator choice, multiple-testing correction, and walk-forward validation. (iii) *A causal mechanism*: a mean-ablation experiment showing the covariance signal is real but dominated by the mean field, and a tail-dependence analysis showing the residual dependence is non-elliptical (upper-tail-independent, radially asymmetric), hence invisible to covariance-based ambiguity sets by construction. (iv) *Practice and theory implications*: a per-region marginal scheduler captures essentially all attainable value, and the evidence yields a precise specification mandate for copula-based DRO.

1.4 Thesis structure

Section 2 reviews the literature and locates the gap. Section 3 develops the optimization model, the falsification test, and the evaluation protocol. Section 4 reports the correlation validation and the experimental results across all region sets, including the robustness battery, mean-ablation, tail-dependence diagnostics, and the Phase 2 copula schedulers. Section 5 interprets the findings against prior work and states limitations. Section 6 concludes and outlines future work.

2 Literature Review

2.1 Carbon-aware computing

Carbon-aware scheduling exploits temporal and spatial variation in grid carbon intensity. In production at Google, Radovanović et al. [17] describe a system which computes day-ahead "virtual capacity curves" that throttle flexible compute during high-carbon hours; measured impact on latency-sensitive workloads is negligible while load shifts measurably toward cleaner hours. Wiesner et al. [20] analyse the potential of temporal shifting across regions and workloads, finding the benefit depends strongly on the grid's marginal mix and the workload's flexibility. Wijayawardana and Chien [21] study cloud-VM scheduling on datacenters whose capacity varies with grid conditions (including carbon-coupled capacity, where a constant carbon budget forces capacity to fall as intensity rises) and document the goodput cost of capacity variation for long-running jobs. This systems literature establishes operational levers (deferral, throttling, capacity coupling) that we adopt as constraints, but it treats future carbon intensity as a point forecast; uncertainty is handled implicitly, by re-planning.

2.2 Distributionally robust optimization

DRO optimizes against the worst-case distribution within an ambiguity set, and the Wasserstein ball around the empirical distribution has emerged as the canonical data-driven choice, with tractable finite-dimensional reformulations [5, 15]. For costs linear in the uncertain vector, type-2 Wasserstein DRO with a Mahalanobis ground metric reduces to a mean term plus a covariance-weighted regularizer, an interpretable "mean plus $\varepsilon \sqrt{x^\top \Sigma x}$" objective solvable as a second-order cone program (SOCP) [2]. Hall et al. [6] apply Wasserstein DRO to carbon-aware data-center scheduling with virtual capacity curves, treating each region's carbon intensity as the uncertain quantity and demonstrating robustness gains over sample-average approximation. The risk functional we target, CVaR_β, is standard for tail-focused evaluation [18]. Beyond second moments, recent work robustifies the *dependence structure* itself: Fan et al. [3] place both the marginals and the *copula* within a Wasserstein ball, the natural home for the non-elliptical structure our Phase 2 copula schedulers target.

2.3 Spatial dependence and its modelling

Cross-region dependence enters a Mahalanobis–Wasserstein model only through the off-diagonal blocks of the covariance matrix. Estimating large covariance matrices from short panels is noisy; shrinkage estimators [16] are the standard remedy and serve here as a robustness arm. Beyond second moments, dependence is fully described by the copula [8, 19]; tail-dependence coefficients quantify joint extremes that correlation cannot see, and the Gaussian copula has zero asymptotic tail dependence for any correlation below one [4], a property we exploit as a diagnostic benchmark. Vine (pair-copula) constructions make high-dimensional non-elliptical dependence tractable [8, 9, 11], with established sequential structure-selection procedures [10], and are the natural candidate for a dependence-aware extension of DRO ambiguity sets. Copula-based scenario generation for stochastic programs is likewise well developed [12].

2.4 The gap

To our knowledge, no prior work *tests* whether spatial correlation of carbon intensity (however strong in the data) translates into out-of-sample robust scheduling value. The systems literature does not model uncertainty explicitly; the DRO literature models uncertainty per region but has not isolated the marginal contribution of *cross-region* dependence; and neither offers a controlled mechanism for *why* spatial structure should or should not matter. This thesis fills that gap with a falsification test, a replicated multi-grid study, and two mechanism diagnostics (mean-ablation and tail dependence).

3 Methodology

3.1 Research design overview

The design is a controlled computational experiment. We hold the optimization machinery fixed and manipulate exactly one object (the cross-region blocks of the covariance supplied to the scheduler) then measure the consequence on held-out tail emissions. Three primary region sets spanning the dependence spectrum (with two earlier panels corroborating) serve as replications; three nested constraint regimes guard against the result being an artifact of a particular feasible-set geometry; a robustness battery (estimators, multiple-testing correction, walk-forward) guards against statistical artifacts; and two further experiments (mean-ablation; tail dependence) identify the mechanism. Throughout, a pre-registration discipline removes analyst degrees of freedom.

3.2 The scheduling problem

Compute is scheduled across R regions over a $T = 24$-hour day. The decision is $x \in \mathbb{R}^{R\times T}_{\geq 0}$, where $x_{r,t}$ is compute power (MW) in region r at hour t. Carbon intensity $\rho_{r,t}$ ($\mathrm{gCO_2eq/kWh}$) is uncertain; realized emissions are linear, $\langle \rho, x\rangle = \sum_{r,t} \rho_{r,t} x_{r,t}$.

3.3 Distributionally robust formulation

We minimize worst-case expected emissions over a type-2 Wasserstein ball $\mathcal{B}_\varepsilon(\hat{\mathbb{P}})$ of radius ε centred on the empirical distribution of the daily carbon field, with Mahalanobis ground metric induced by the empirical covariance $\hat{\Sigma} \in \mathbb{R}^{RT\times RT}$:

$$\min_{x\in\mathcal{X}} \sup_{\mathbb{Q}\in\mathcal{B}_\varepsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{Q}}\big[\langle \rho, x\rangle\big] = \min_{x\in\mathcal{X}} \langle \bar{\rho}, x\rangle + \varepsilon\sqrt{x^\top \hat{\Sigma}\, x}, \tag{1}$$

by Wasserstein duality for linear costs [2, 5], where $\bar{\rho}$ is the empirical mean field. With the Cholesky factorization $\hat{\Sigma} = LL^\top$ the penalty becomes $\varepsilon\|L^\top \operatorname{vec}(x)\|_2$. Introducing an epigraph variable η (distinct from the hour index t), (1) is the second-order cone program

$$\min_{x\in\mathcal{X},\,\eta\geq 0} \langle\bar{\rho}, x\rangle + \varepsilon\,\eta \quad \text{s.t.} \quad \left\|L^\top \operatorname{vec}(x)\right\|_2 \leq \eta, \tag{2}$$

solved to global optimality (CLARABEL). The radius ε interpolates from the deterministic LP ($\varepsilon = 0$) to increasingly conservative schedules.

Where spatial correlation enters. Partitioning $\hat{\Sigma}$ into $T \times T$ blocks Σ_{rs} by region pair, the quadratic form splits as

$$x^\top\hat{\Sigma}x = \sum_r x_r^\top \Sigma_{rr} x_r + \sum_{r\neq s} x_r^\top \Sigma_{rs} x_s, \tag{3}$$

so the off-diagonal (spatial) blocks are the *only* channel by which cross-region dependence can alter the schedule. The experiment manipulates exactly these blocks.

3.4 Feasible set and constraint regimes

The feasible set $\mathcal{X}$ collects the linear constraints below; introducing an auxiliary flexible-load variable $x^{\mathrm{f}} \in \mathbb{R}^{R\times T}_{\geq 0}$, the scheduler solves (1) over

$$\begin{aligned}
&0 \leq x_{r,t} \leq \bar{x}_{r,t}, && \forall r,t && \text{(C0: capacity)} && \text{(4a)}\\
&x_{r,t} = \alpha_r W_r\, p_{r,t} + x^{\mathrm{f}}_{r,t}, && \forall r,t && \text{(C2a: flex/inflex split)} && \text{(4b)}\\
&\textstyle\sum_t x^{\mathrm{f}}_{r,t} = (1-\alpha_r)\, W_r, && \forall r && \text{(C2b: work conservation)} && \text{(4c)}\\
&|x_{r,t} - x_{r,t-1}| \leq \Delta_r, && \forall r,t && \text{(C3: ramp)} && \text{(4d)}\\
&\textstyle\sum_{t\in\mathcal{W}} x^{\mathrm{f}}_{r,t} \geq \gamma\,(1-\alpha_r)\, W_r, && \forall r && \text{(3a: deferral deadline)} && \text{(4e)}\\
&\pi_{r,t}\, x_{r,t} \leq \bar{P}, && \forall r,t && \text{(3b: thermal/PUE)} && \text{(4f)}\\
&\textstyle\sum_{r,t} \bar{\rho}_{r,t}\, x_{r,t} \leq B && && \text{(3d: carbon budget)} && \text{(4g)}
\end{aligned}$$

where W_r is region r's daily work (utilization fixed at 80%, $W_r = 960\,\text{MWh}$); $\alpha_r \in \{0.30, 0.50, 0.75\}$ is the inflexible fraction following the fixed intraday shape $p_{r,t}$ ($\sum_t p_{r,t} = 1$); $\Delta_r = 15$ MW/h is the ramp limit; $\mathcal{W} = [0, 7]$ and $\gamma = 0.20$ set the deferral deadline; and $\pi_{r,t} = \text{PUE}_0 + \kappa \max(\mathcal{T}_{r,t} - \mathcal{T}_{\text{set}}, 0)$ is the temperature-coupled power-usage multiplier (data, hence a constant per cell), capped at effective power $\bar{P}$. Every constraint is linear, so $\mathcal{X}$ is polyhedral and the program (1) remains an SOCP. (We follow the source labelling: C0–C3 are the base operational constraints and the **3**· series is adapted from the SoCC'25 formulation [21]; a further aggregate hourly cap $\sum_r x_{r,t} \le p_{\max}$ is implemented but inactive in the reported regimes.)

The capacity ceiling is constant ($\bar{x}_{r,t} = 50$ MW) except under constraint **3c**, the carbon-coupled variable capacity adapted from Wijayawardana and Chien [21], which replaces it with

$$\bar{x}_{r,t} = x_{\min} + (x_{\max} - x_{\min})\,\mathrm{CFE}_{r,t}/100, \tag{5}$$

where $\mathrm{CFE}_{r,t}$ is the training-mean carbon-free-energy share, capacity falls as the grid gets dirtier. The bounds $(x_{\min}, x_{\max}) = (50, 75)$ were calibrated on training data to a loosely binding ("Goldilocks") regime: the constraint binds on 28–37% of region-hours, leaving positive slack and capacity margin, so it shapes the schedule without freezing it.

Three nested regimes report results as a function of operational tightness: **R3** (reference) imposes C0 + C2 + C3; **R1** (lean) adds the deferral deadline 3a; and **R2** adds the thermal ceiling 3b (climate cases) or the variable-capacity ceiling 3c (interconnection cases). The carbon budget 3d is available but inactive in the reported regimes. Each regime is solved as a *complete* model with all of its constraints active jointly; the nesting probes the null's robustness to operational tightness, and is not a test of any single constraint's effect in isolation.

3.5 Model validation on a small instance

Before running the falsification experiment we check that the optimization model itself behaves correctly. The full model is validated *as a whole*, with all constraints of (4) active simultaneously, on a small, human-checkable *two-region* instance (two regions, eight hours, 10 MWh per-cell capacity, $W = 30$ MWh each), rather than each constraint in isolation: correctness is a property of the constraints' *interaction*, and a set that every constraint satisfies alone can still be jointly binding in unexpected ways. We use two regions specifically so the *cross-region* interaction is exercised: a shared aggregate hourly cap $\sum_r x_{r,t} \le P_{\mathrm{agg}}$ couples the regions, which must then compete for the same clean hours. We fix an analyst-anticipated outcome, perturb one input at a time, and verify the solved schedule moves the anticipated way (Figure 1). At baseline the two regions' clean windows overlap, and the aggregate cap forces them to split the contested hours (panel A). Making region 1's clean window dirty drives it to cede those hours to region 2 and spread to its own next-cleanest hours, with each region's work still conserved (panel B). Tightening the shared cap removes joint headroom, so both regions spread further in time rather than co-locating in the clean window (panel C). All three responses match the hand-anticipated solution and conserve work per region to machine precision. This model-level check is complementary to, and distinct from, the software test suite (Section 3.10, which pins component behaviour such as feasible-set equivalence) and the falsification experiment below (which probes a scientific hypothesis, not model correctness).

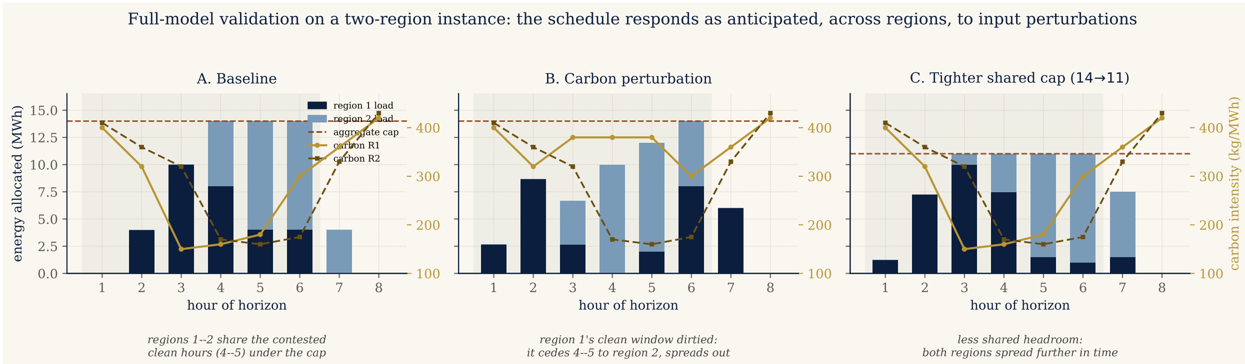

Figure 1: Full-model validation on a two-region instance. The complete feasible set (capacity, work conservation, ramp, deferral deadline) plus a shared aggregate hourly cap is solved jointly; stacked bars show how the two regions share each hour, the dashed line is the aggregate cap, the curves are each region's carbon field, the green band the deadline window. **A:** baseline, the regions split the contested clean hours (4–5) under the cap. **B:** after dirtying region 1's clean window, it cedes those hours to region 2 and spreads out. **C:** after tightening the shared cap, both regions spread further in time. Each response is the one an analyst anticipates, confirming the model encodes the intended cross-region behaviour.

3.6 The falsification test: shuffled marginals

The scheduler is fitted twice, with covariances differing in exactly one respect:

$$\left(\hat{\Sigma}^{\text{shuf}}\right)_{rs} = \begin{cases} \Sigma_{rr}, & r = s, \\ \mathbf{0}, & r \neq s, \end{cases} \tag{6}$$

preserving every region's marginal distribution and within-region autocorrelation while destroying only cross-region dependence. Each fitted schedule x is evaluated by the out-of-sample conditional value-at-risk of daily emissions $e_i = \langle \rho_i, x \rangle$ realized on test day i; CVaR is a coherent risk measure [13] standard in stochastic programming [14]. We use the Rockafellar–Uryasev form [18]

$$\text{CVaR}_\beta(e) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \tfrac{1}{1-\beta}\, \mathbb{E}\big[(e-\tau)_+\big] \right\}, \qquad \beta = 0.95, \tag{7}$$

estimated as the empirical mean of the worst 5% of test days. The *spatial gap* is the difference between the two arms,

$$\text{gap} = \text{CVaR}_{0.95}\left(e^{\text{shuf}}\right) - \text{CVaR}_{0.95}\left(e^{\text{joint}}\right), \tag{8}$$

positive when the joint covariance helps. The test is a falsification: it is built to detect a spatial benefit if one exists, so a null is informative. Figure 2 sketches the pipeline and Algorithm 1 gives the full protocol.

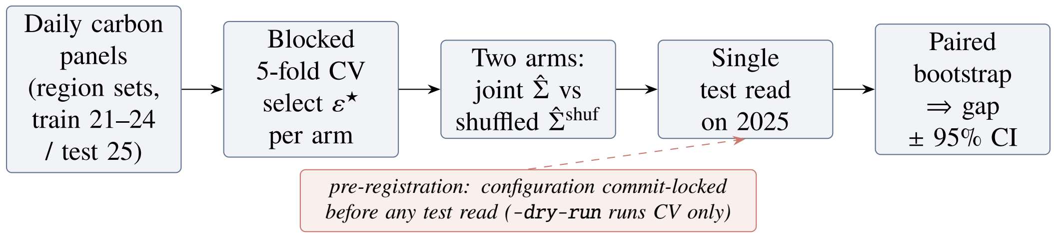

Figure 2: The pre-registered falsification pipeline. Cross-validation and schedule fitting touch only the training years; the 2025 test set is read once, after the configuration is commit-locked, and the spatial gap is read off a paired day-bootstrap.

Algorithm 1 Shuffled-marginals experiment (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); regimes {R3, R1, R2}; α-levels $\{0.30, 0.50, 0.75\}$; grid $\mathcal{E} = \{0, 0.1, 1, 10, 100, 1000\}$
for each regime, each α, each arm $\Sigma \in \{\text{joint}, \text{shuf}\}$ **do**
    for each $\varepsilon \in \mathcal{E}$ **do** ▷ blocked 5-fold time-series CV
        for each contiguous fold k **do**
            fit $\bar{\rho}$, L on training folds; solve SOCP (1); record validation $\text{CVaR}_{0.95}$
    $\varepsilon^\star \leftarrow \arg\min_\varepsilon$ mean validation CVaR ▷ per arm, independently
    refit on full training set at $\varepsilon^\star$; evaluate *once* on 2025 ▷ single test read
paired bootstrap (1000 resamples, fixed seed) on test days $\Rightarrow$ 95% CI for the gap (8)
Pre-registration: configuration commit-locked before any test read; `-dry-run` gate runs CV only

3.7 Data

Carbon intensity is hourly lifecycle intensity from Electricity Maps (academic licence), 2021–2025, 43,824 hourly observations per zone; panels are built on a common local clock per region set, trained on 2021–2024 and tested once on 2025. Three primary US/Canada region sets span the dependence spectrum: **(1)** the *US West*, California, Nevada and Phoenix (CISO, BANC, LDWP, NEVP, AZPS); **(2)** an *Ontario–Eastern belt* (CA-ON, NYISO, MISO, PJM); and **(3)** an *engineered heterogeneous portfolio* (CISO (solar), ERCOT (wind), BPAT (hydro), spanning *different* interconnections) deliberately chosen so clean periods do *not* align: the adversarial best case for diversification. This spread is deliberate on both ends of the spectrum. The two correlated, within-interconnection grids are the *best case for the spatial hypothesis*, the most cross-region correlation a covariance model could possibly exploit, and double as a stress test rather than a realistic deployment (no operator clusters data centers within one interconnection). A realistic, globally dispersed fleet looks instead like the heterogeneous portfolio, low and mixed correlation, so the practically relevant case is the one where diversification *should* help most. The null holding at both ends is therefore stronger than a null on either alone. Two earlier panels corroborate: an *Iberia–France* European set (ES, PT, FR), where residual correlation collapses to the diurnal cycle, and the California/Nevada subset of the US West before Phoenix was added at the supervisor's request. Temperature for constraint 3b is hourly ERA5 2 m air temperature (Open-Meteo), one load-weighted point per zone. The licensed raw data is never committed; derived summary tables are archived in the repository for traceability.

3.8 Validation and mechanism diagnostics

Correlation validation (RQ1). Raw Pearson correlation conflates co-movement with a shared diurnal cycle. We therefore report correlations of *residuals* after removing each zone's hour-of-day mean (local clock), with overlay plots of standardized intensity for summer and winter weeks.

Robustness battery. Every experiment is re-run with (i) Ledoit–Wolf shrinkage covariance [16]; (ii) covariance estimated from seasonal (monthly-mean) residuals; (iii) covariance from AR(1) residuals. A pre-registered agreement rule counts a spatial effect only if positive and sign-stable across both residual estimators. Bootstrap p-values across all cells receive Benjamini–Hochberg correction at $q = 0.05$ [1]; a walk-forward refit (train 2021–2023, test 2024) checks temporal stability; a log-denser ε grid checks grid sensitivity.

Mean-ablation (RQ3, causal). To test whether the mean field masks the covariance, the mean used for *scheduling* is transformed while evaluation stays on real emissions: LEVEL equalizes regional time-averages (keeping hour-of-day shape); FLAT sets a constant mean, making the mean term constant under fixed demand, a covariance-only world in which any joint-vs-shuf difference is attributable purely to the spatial blocks.

Tail dependence (RQ3, structural). For each region pair we estimate the upper and lower tail-dependence coefficients $\chi_{\mathrm{U}}(q) = \Pr(U > q, V > q)/(1 - q)$ and $\chi_{\mathrm{L}}(p) = \Pr(U < p, V < p)/p$ on rank pseudo-observations of residual intensity, comparing χ_{U} at $q = 0.95$ with the Gaussian-copula value at the matched correlation. Since the Gaussian copula, the dependence family a covariance ball can represent, is asymptotically tail-independent [4], a positive empirical excess would be exploitable tail structure invisible to the DRO; and since elliptical copulas force $\chi_{\mathrm{L}} = \chi_{\mathrm{U}}$, radial asymmetry diagnoses non-elliptical dependence.

3.9 Phase 2: copula dependence models and a mean-dominance bound

The covariance test answers whether *elliptical* dependence helps. Because the tail diagnostic flags *non-elliptical* structure ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), a covariance ball cannot by construction represent it, so a natural objection to the Phase 1 null is "you used the wrong dependence object." Phase 2 forecloses that objection by replacing the covariance ball with copula models that *can* represent the structure, and re-running the same falsification.

Copula-coupled resampling. We generalize the shuffled-marginals test from second moments to the full copula while preserving every region's empirical marginal. Each region r keeps its pool of whole daily profiles $\{\rho_r^d \in \mathbb{R}^T\}$; a scenario is generated by drawing a copula sample $u \in [0,1]^R$ and, for each region, selecting the training day whose daily-mean intensity has rank u_r (low rank = clean day), then taking that day's full profile. The copula thus governs *only* the cross-region coupling, exactly as the shuffled covariance did. Three nested fitted models span the dependence hierarchy (a fourth, the comonotone upper bound, is added in Section 4.6 as the ceiling): **(i)** the *independence* copula (regions decoupled, the Phase 1 shuffled arm); **(ii)** the *Gaussian* copula at the empirical rank-correlation (elliptical, what the covariance ball can see); and **(iii)** an exchangeable *Clayton* copula, with lower-tail dependence $\lambda_L = 2^{-1/\theta}$ and zero upper-tail dependence [8], fitted by matching mean pairwise Kendall's τ ($\theta = 2\tau/(1-\tau)$), the object built for the $\chi_\mathrm{L} > \chi_\mathrm{U}$ finding. Clayton is sampled by Marshall–Olkin frailty; no parametric marginal model is introduced. The exchangeable Clayton imposes a single θ and so cannot reproduce the *pair-level* heterogeneity of the tail diagnostic (Table 6); a regular-vine copula with edge-specific families would. We do not fit one, for a principled reason: the comonotone arm below realizes the supremum over *all* copulas, vines included, so it upper-bounds whatever any heterogeneous-dependence model could achieve. If the maximal-coupling arm recovers no material value, no vine can.

Copula CVaR scheduler. For each model we draw $S = 1000$ scenarios $\{\rho^s\}$ and solve the sample-average CVaR program in Rockafellar–Uryasev form [18],

$$\min_{x\in\mathcal{X},\,\tau,\,z\geq 0} \tau + \frac{1}{(1-\beta)S}\sum_{s=1}^{S} z_s \quad \text{s.t.} \quad z_s \geq \langle \rho^s, x\rangle - \tau, \tag{9}$$

over the *same* feasible set $\mathcal{X}$ (4) and regimes as Phase 1, so the only thing that varies across arms is the dependence model. Each schedule is evaluated once on the 2025 panel by out-of-sample $\mathrm{CVaR}_{0.95}$; the reported copula gaps are $\mathrm{CVaR}(\text{indep}) - \mathrm{CVaR}(\text{Gaussian})$, $\mathrm{CVaR}(\text{indep}) - \mathrm{CVaR}(\text{Clayton})$, and the asymmetry increment $\mathrm{CVaR}(\text{Gaussian}) - \mathrm{CVaR}(\text{Clayton})$, each with a paired day-bootstrap CI. Algorithm 2 summarizes the procedure.

Why the dependence model is second-order. Two facts make the dependence model second-order to the mean field. The first is a decomposition: by the translation invariance of CVaR, writing

Algorithm 2 Copula-scenario CVaR scheduling (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); dependence models {independence, Gaussian, Clayton, comonotone}; regimes; α-levels; scenarios $S = 1000$, fixed seed
for each dependence model C **do**
  fit C on training daily summaries (rank pseudo-observations); for Clayton, $\theta = 2\tau/(1-\tau)$ from mean Kendall τ
  for each regime, each α **do**
    draw S copula samples $u^s \in [0,1]^R$ ▷ indep./Gaussian/Clayton frailty/comonotone
    map each region's u^s_r to the training day of that daily-summary rank; take its full profile $\Rightarrow$ scenario $\rho^s \in \mathbb{R}^{R\times T}$
      ▷ copula-coupled resampling: marginals exact, only cross-region coupling changes
    solve the CVaR-SAA program (9) over $\mathcal{X} \Rightarrow$ schedule x
    evaluate *once* on 2025 $\Rightarrow$ out-of-sample $\mathrm{CVaR}_{0.95}$
copula gaps vs. independence + paired day-bootstrap CIs; comonotone realizes $\sup_C$, the ceiling Λ

the carbon field as mean plus zero-mean residual $\rho = \bar{\rho} + \xi$ and noting $\langle\bar{\rho}, x\rangle$ is deterministic,

$$\mathrm{CVaR}_\beta\big(\langle\rho, x\rangle\big) = \underbrace{\langle\bar{\rho}, x\rangle}_{\text{dependence-independent}} + \mathrm{CVaR}_\beta\big(\langle\xi, x\rangle\big), \tag{10}$$

so for a *fixed* schedule the dependence model moves only the residual term. The second is an a-priori bound on how far it can move the *optimal* value.

Proposition 1 (Spatial gap control, SOCP)**.** *Let* $\mathrm{OPT}(\Sigma)$ *be the optimal value of the SOCP* (2) *for a covariance* Σ*, and let* Σ^{shuf} *be its block-diagonal (cross-region-zeroed) counterpart with* $\Sigma^{\mathrm{off}} = \Sigma - \Sigma^{\mathrm{shuf}}$*. Then the spatial gap obeys*

$$\big|\mathrm{OPT}(\Sigma) - \mathrm{OPT}(\Sigma^{\mathrm{shuf}})\big| \;\le\; \varepsilon\left(\max_{x\in\mathcal{X}}\|x\|_2\right)\big\|\Sigma^{\mathrm{off}}\big\|_2^{1/2}, \tag{11}$$

which vanishes as $\varepsilon \to 0$ *(no robustification) or* $\Sigma^{\mathrm{off}} \to 0$ *(no cross-region structure).*

Proof. The two programs share the feasible set $\mathcal{X}$ and differ only in the conic penalty. For fixed x, using $|\sqrt{a} - \sqrt{b}| \le \sqrt{|a-b|}$ and the Rayleigh bound $|x^\top\Sigma^{\mathrm{off}}x| \le \|\Sigma^{\mathrm{off}}\|_2\|x\|_2^2$,

$$\left|\varepsilon\sqrt{x^\top\Sigma x} - \varepsilon\sqrt{x^\top\Sigma^{\mathrm{shuf}}x}\right| \le \varepsilon\sqrt{|x^\top\Sigma^{\mathrm{off}}x|} \le \varepsilon\|x\|_2\|\Sigma^{\mathrm{off}}\|_2^{1/2}.$$

Since the infimum is 1-Lipschitz in the sup-norm of the objective, two objectives that differ pointwise by at most δ over a common feasible set have optimal values differing by at most δ. The feasible set $\mathcal{X}$ is a bounded polytope (the capacity bounds C0), hence compact, so $\max_{x\in\mathcal{X}}\|x\|_2$ is attained; taking that maximum gives (11). □

The bound is a conservative *certificate*, not a tight prediction. Evaluated on each grid's training covariance (with $\varepsilon^\star = 1$, the a-priori feasible bound $\max_x \|x\|_2 = \bar{x}\sqrt{R\,T\,u}$ where $\bar{x}$ is the per-cell cap and u the utilization, and $\|\Sigma^{\text{off}}\|_2$ the spectral norm of the destroyed cross-region blocks), its right-hand side is 7.4% (Eastern US–Canada), 9.6% (Diversified) and 12.7% (Western US) of $\text{CVaR}_{0.95}$; the realized gaps (Table 1) never exceed 0.23%, two orders of magnitude smaller. The theorem therefore caps the spatial gap at the few-percent scale *before any experiment is run*, and the experiments then pin it far below that cap. What the bound buys is not tightness but structure: the spatial gap is governed by ε and the cross-region block norm, and is forced to zero in the no-robustification and no-cross-structure limits. Most of the remaining slack lives in the feasible-diameter factor $\max_x \|x\|_2$, an a-priori worst case; substituting the realized optimal $\|x^\star\|_2$ would tighten the certificate considerably, at the cost of making it post-hoc rather than a-priori. We keep the a-priori form deliberately, since its purpose is to bound the gap *before* the experiment, not to predict it. The *tight* ceiling is measured, not bounded: the comonotone arm of Section 4.6 realizes the supremum over copulas, and across scenario seeds its gain is centred on zero with $|\Lambda| \lesssim 0.1\%$ of $\text{CVaR}_{0.95}$, at the sampling-noise floor. The mean-ablation of Section 4.4 bounds the same quantity from a second, independent angle. Both put Λ an order of magnitude below the schedule's mean-driven value; hence the mean field pins the schedule and the dependence model, elliptical or not, is immaterial. The prediction that even Clayton cannot recover material value is confirmed in Section 4.6.

Computational considerations. Each instance is small: the decision vector has $RT \leq 5 \times 24 = 120$ entries and the feasible set adds $O(RT)$ linear constraints, so the SOCP (2) solves in 0.19–0.26 s and the CVaR-SAA LP (9), which appends $S = 1000$ epigraph variables and constraints, in 0.9–1.3 s (median over five repetitions, CLARABEL/HiGHS, commodity laptop). The cost is in the *number* of instances, not their size: one region-set experiment solves 3 regimes $\times$ 3 α $\times$ 2 arms $\times$ ($6\,\varepsilon \times 5$ folds + 1 test read) = 558 SOCPs, and the full Phase 1 battery (region sets $\times$ {sample, Ledoit–Wolf, seasonal, AR(1), two ablations, walk-forward, fine grid}) is on the order of 10^4 SOCP solves; Phase 2 adds $3 \times 3 \times 3 \times 4 = 108$ CVaR-SAA LPs. Total wall-clock for the entire study is under two hours on one core, so reproduction is cheap.

3.10 Reproducibility

All code is version-controlled (private repository, continuous-integration test suite, 175 unit tests, including a feasible-set equivalence test pinning the Phase 2 CVaR scheduler to the Phase 1 constraint set); experiment configurations are commit-locked before test-set evaluation; bootstrap and scenario seeds are fixed; and every number reported in Sections 4–5 traces to an archived

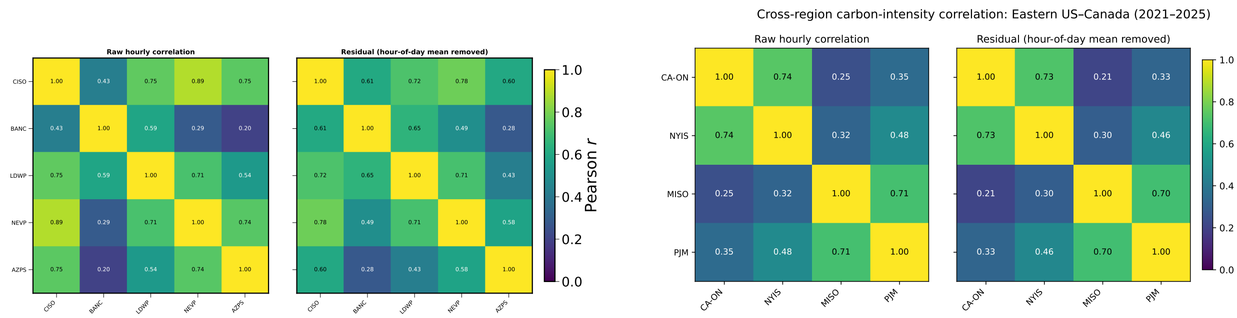

Figure 3: Residual (hour-of-day-removed) cross-region correlation of hourly carbon intensity, 2021–2025. Left: US West (CISO, BANC, LDWP, NEVP, AZPS). Right: Ontario–Eastern belt (CA-ON, NYISO, MISO, PJM). Spatial correlation is strong and genuine in both, the assumption motivating spatial DRO is empirically valid.

summary table. Solvers: CLARABEL/ECOS for the SOCP; HiGHS for the CVaR-SAA linear programs and deterministic LP baselines.

4 Results

4.1 The spatial assumption is valid (RQ1)

The premise holds. After removing each zone's hour-of-day mean, cross-region correlations of carbon intensity in the two electrically coupled US/Canada cases are strong and survive de-seasonalization: in the US West, CISO–NEVP 0.78 and CISO–LDWP 0.72 (shared solar resource and weather across WECC); in the Ontario–Eastern belt, CA-ON–NYISO 0.73 and MISO–PJM 0.70 (weather fronts traversing the Eastern Interconnection). The engineered portfolio is the low, *heterogeneous* end, not a zero-correlation grid: its residual pairs span 0.00 (ERCOT–BPAT), 0.29 (CISO–ERCOT) and 0.37 (CISO–BPAT), mean $\bar{r} \approx 0.2$, a deliberately mixed generation profile. Such partly-low, heterogeneous correlation is *diversifiable* (it drives the largest mean-ablation signal, Section 4.4), unlike the US West's strong but *uniform*, common-mode correlation. Figure 3 shows the residual correlation matrices; overlay plots of standardized intensity (archived with the code) confirm visible co-movement in summer and winter weeks for the coupled cases. Together with the corroborating panels, the California/Nevada subset (0.65–0.78 within-state) and Iberia–France, the cases span the dependence spectrum from ≈ 0 to ≈ 0.8, so whatever the scheduling result, it cannot be attributed to an absence of correlation in the data.

Figure 4 places the three grids geographically and draws each within-grid residual correlation as a weighted edge. The picture matches the physics: the US West forms one tightly-coupled WECC cluster, the Ontario–Eastern belt another along the Eastern Interconnection, while the engineered

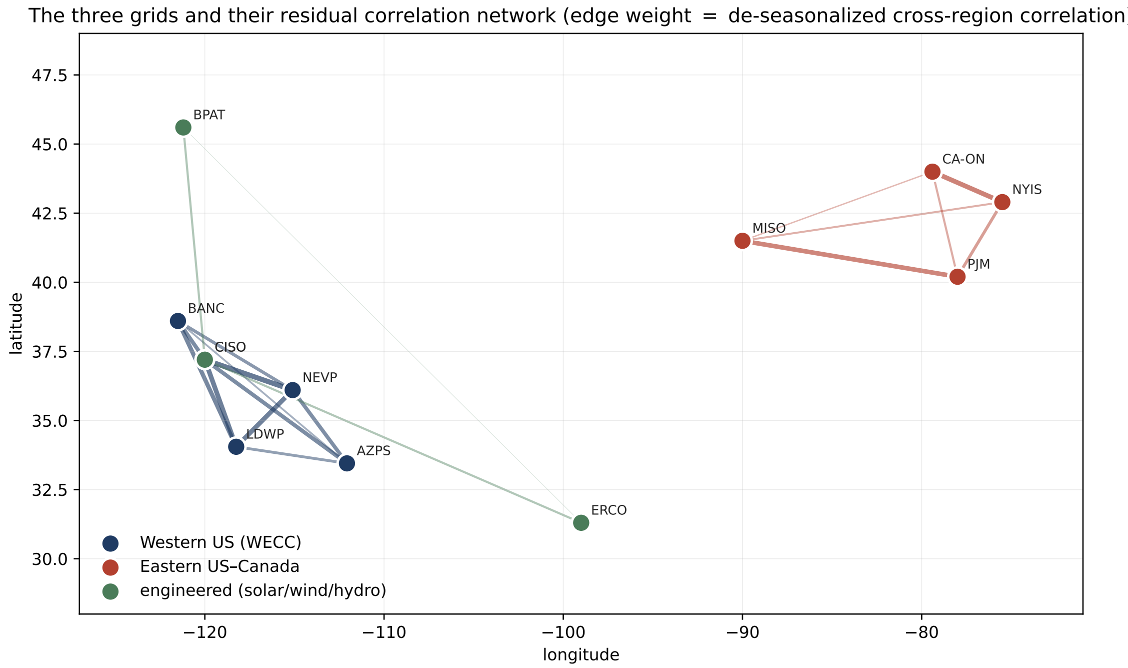

Figure 4: The three grids and their residual correlation network. Nodes are Electricity Maps zones at their geographic centroids; edge weight is the de-seasonalized cross-region correlation. Tight clusters (US West, Ontario–Eastern belt) contrast with the deliberately diversified engineered portfolio.

portfolio spans the continent with mixed edges, one near-zero and two moderate, by construction.

4.2 The spatial gap is null across the spectrum (RQ2)

Table 1 reports the spatial gap (8) for the three pre-registered US/Canada cases: nine cells each (three constraint regimes × three α-levels), CV-selected $\varepsilon^\star$, single test read on 2025. The gap never exceeds a few tenths of one percent of test $\mathrm{CVaR}_{0.95}$ in either direction. Where the bootstrap calls a cell "detectable" the effect is genuine but economically negligible (on the order of a few hundred $\mathrm{gCO_2eq/kWh}$ out of a CVaR of 1.5×10^6) and, decisively, its *sign* flips across adjacent α-levels within the same regime. A consistently exploitable spatial structure would not reverse direction under a change of the flexible-load fraction; these reversals are the signature of optimization at the boundary of materiality, not of a usable effect. (We confirmed the detectable cells are not numerical: re-solving with CLARABEL and SCS reproduces the gap to four significant figures, so the sign instability is in the data, not the solver.) The California/Nevada panel agrees (gaps in $[-0.012, +0.058]\%$, DRO active throughout).

Crucially, this is a null of the *strong* kind. In all three pre-registered cases cross-validation *chooses* a non-trivial ambiguity radius: $\varepsilon^\star = 1$ in every one of the 27 cells for Eastern US–Canada

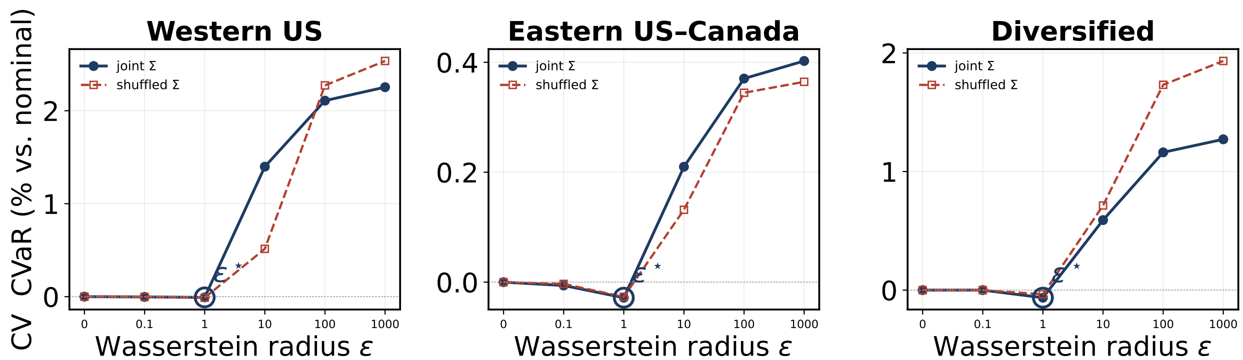

Figure 5: The DRO genuinely engages. Blocked 5-fold cross-validation $\mathrm{CVaR}_{0.95}$ (relative to the nominal $\varepsilon = 0$ problem) against the Wasserstein radius ε, for the joint (solid) and shuffled (dashed) covariances. Cross-validation selects an interior $\varepsilon^\star$ (circled), not $\varepsilon = 0$, so the robustification is on; the two curves coincide, so the spatial covariance changes nothing. Large ε is over-conservative.

and Diversified, and in 25 of 27 for Western US. The DRO is therefore genuinely engaged (it is paying for robustness, not collapsing to the nominal problem) and the two arms still coincide. A null in which CV had selected $\varepsilon^\star = 0$ would be uninformative (robustification switched off, so the arms must match by construction); here the robustification is demonstrably on and the spatial covariance still buys nothing. This is the first regime in the project where the DRO is consistently active, which is precisely what makes the null informative rather than vacuous. Figure 5 shows the cross-validation curves directly: validation $\mathrm{CVaR}_{0.95}$ dips to an interior minimum at a non-trivial $\varepsilon^\star$ before rising as over-conservatism sets in, and the joint and shuffled curves lie on top of one another.

The Iberia–France panel is the instructive contrast and the project's historical starting point. There the residual correlation is near zero, and CV does not even consistently engage the DRO: $\varepsilon^\star$ flickers between 0, 0.1, 1 and 10 across cells, collapsing to 0 in several. This is itself consistent with the thesis (where there is no cross-region structure, the machinery sees nothing worth robustifying) but it makes the Iberian null the *weak* kind. We therefore retain it only as low-correlation, real-grid external validity, and rest the central claim on the US/Canada cases where the DRO is unambiguously on. The engineered Diversified portfolio supplies the DRO-active low-correlation null ($\varepsilon^\star = 1$ throughout, still no value) that Iberia cannot.

Three readings of Table 1 matter. First, in the adversarial diversification case (Diversified) the joint covariance actually *hurts* at low α (up to -0.23%): with no true cross-structure, the joint estimator fits noise that the block-diagonal arm is spared. Second, in the strongly correlated belt (Eastern US–Canada) the detectable cells flip sign between $\alpha = 0.30$ and $\alpha = 0.75$ within the same regime. Third, the US West (strongest correlation of all) produces no detectable cell beyond one of magnitude $+3 \times 10^{-6}\%$. Figure 6A plots all 27 cells against each case's mean residual correlation:

Table 1: Spatial gap (% of test $\mathrm{CVaR}_{0.95}$; positive = joint covariance better) for the three US/Canada cases. R3 = reference regime, R1 = +deadline, R2 = +variable carbon-coupled capacity. "det." = bootstrap 95% CI excludes zero.

		Diversified ($\bar{r} \approx 0.2$)		Eastern US–Canada ($\bar{r} \approx 0.45$)		Western US ($\bar{r} \approx 0.65$)	
Regime	α	gap %	det.	gap %	det.	gap %	det.
R3	0.30	−0.233	✓	−0.022	✓	+0.043	
R3	0.50	−0.177	✓	−0.002		+0.032	
R3	0.75	+0.001		+0.034	✓	−0.005	
R1	0.30	−0.233	✓	−0.022	✓	+0.014	
R1	0.50	−0.177	✓	−0.002		+0.028	
R1	0.75	+0.001		+0.034	✓	+0.002	
R2	0.30	−0.211	✓	+0.038	✓	+0.008	
R2	0.50	−0.154	✓	+0.030	✓	+0.005	
R2	0.75	+0.062	✓	+0.009		+0.000	✓

the cloud stays on zero across the whole correlation range. **Validity of the spatial assumption does not imply value from it.**

The null is visible in the schedules themselves. Figure 7 overlays the optimal load for the joint and shuffled covariances on each region's mean carbon field: the two schedules are indistinguishable to the eye, and both place compute in the cleaner hours subject to the deadline and ramp constraints. Destroying the cross-region covariance changes essentially nothing about what the scheduler does.

4.3 The null survives a pre-registered robustness battery (RQ2)

Table 4 summarizes. Re-estimating the covariance with Ledoit–Wolf shrinkage, seasonal residuals, or AR(1) residuals never produces a spatial effect that passes the pre-registered agreement rule (positive and sign-stable across both residual estimators). The largest battery value anywhere, +0.179% (AR(1), Eastern US–Canada, $\alpha = 0.30$), appears *only* under AR(1) and not under the seasonal estimator, so the rule rejects it. Benjamini–Hochberg correction at $q = 0.05$ across all 144 non-ablation cells (four grid-runs, the three primary grids plus the locked Task C replication, each at 3 regimes $\times$ 3 α = 9 cells under the four covariance estimators {sample, Ledoit–Wolf, seasonal, AR(1)}: $4 \times 9 \times 4 = 144$) leaves that same already-rejected cell as the largest surviving positive gap; nothing material survives both filters. A walk-forward refit (train 2021–2023, test 2024) reproduces the null out of time: maximum absolute gap 0.036% (Eastern US–Canada), 0.028% (Western US), 0.217% (Diversified, again negative). An 11-point log-denser ε grid changes no gap at the reported precision. Finally, a *constraint-tightness* sensitivity directly probes the most plausible route to spatial value: tripling the binding tightness of the ramp limit ($\Delta_r = 5$ instead of 15 MW/h) shrinks the feasible set so the schedule has far less freedom to track the mean field, yet the null is unmoved,

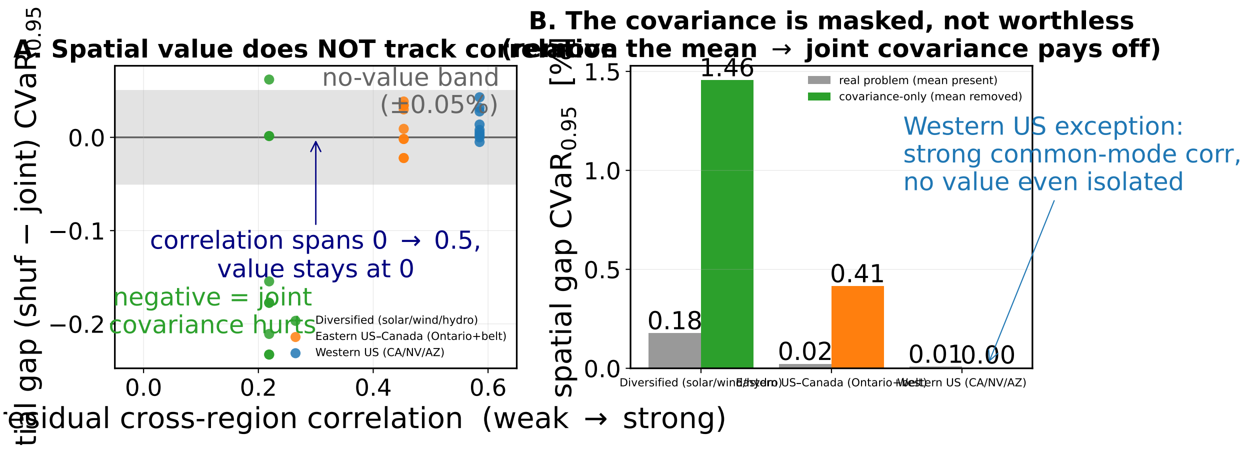

Figure 6: The Phase 1 finding. **A:** spatial gap vs. mean residual cross-region correlation; value does not track correlation anywhere on the spectrum. **B:** mean-ablation; in the covariance-only world (constant scheduling mean) the joint covariance pays off by up to +1.46%, proving the spatial signal is real but *masked* by the mean carbon field, not absent.

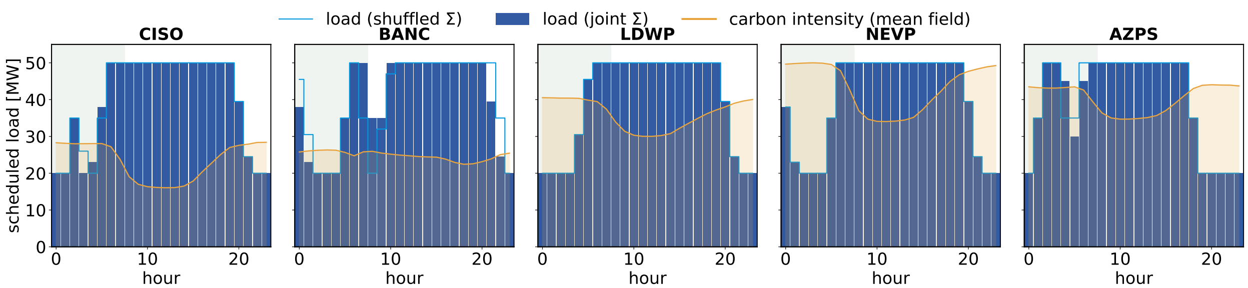

Figure 7: The spatial null, seen directly (Western US, regime R3, $\alpha = 0.5$, $\varepsilon^{\star} = 1$). Bars are the optimal load under the joint covariance; the red step is the load under the shuffled (block-diagonal) covariance. The two coincide. The gold curve is each region's mean carbon intensity; the green band marks the deadline window $[0, 7]$.

with maximum absolute gap 0.028% (Western US), 0.044% (Eastern US–Canada) and 0.169% (Diversified), all sign-flipping and with $\varepsilon^{\star} = 1$ retained. The covariance does not begin to pay even when the mean's pull is mechanically curtailed. A complementary *utilization* sweep makes the same point through the workload: raising daily utilization from the baseline 80% to 95% packs the facility near capacity and strips out almost all scheduling freedom, yet the gap only *shrinks* (maximum absolute gap 0.017% for Western US, 0.048% for Eastern US–Canada, 0.071% for Diversified), and at a loose 50% it stays immaterial ($\leq 0.25\%$); $\varepsilon^{\star} = 1$ throughout. Across the full $\{50, 80, 95\}\%$ range the null is unmoved. Figure 8 collects the covariance-estimator and temporal battery: across all six arms and three grids, every spatial gap sits inside the no-value band.

Tail level: the null holds from $\mathrm{CVaR}_{0.90}$ **to** $\mathrm{CVaR}_{0.99}$**.** The primary read fixes the tail at $\beta = 0.95$. Because the residual co-movement lives in the *clean* tail (Section 4.5), the natural worry is that spatial value might surface only at a deeper risk level. Re-running the full pipeline (cross-validated ε, single 2025 read) at $\beta \in \{0.90, 0.95, 0.99\}$ rejects this (Table 2). The DRO stays engaged ($\varepsilon^\star = 1$ almost everywhere) and the gap stays immaterial at every level. Probing the deepest tail widens the magnitudes slightly but never to materiality: the largest *positive* gap anywhere is +0.14% (Western US, $\beta = 0.99$), still a third of the 0.4% margin and not sign-stable, while in the adversarial Diversified grid the joint covariance becomes *more* negative at $\beta = 0.99$ (−0.36%, i.e. it hurts), exactly as the noise-fitting account predicts. Deepening the tail, the regime where elliptical-blind tail dependence should matter most, does not turn the covariance on. The $\beta = 0.99$ estimates are necessarily noisier, since the tail thins to a handful of days: the per-cell paired-bootstrap standard error rises to 0.15% for Western US (against ≤ 0.04% at $\beta = 0.90$), so that grid's largest $\beta = 0.99$ point estimate (+0.14%) is itself within one standard error of zero. The apparent widening at the deepest tail is sampling noise, not emerging spatial value. This sweep is reported as a descriptive robustness check, *outside* the pre-registered 144-cell FDR family of Table 4; for completeness, no sweep cell produces a gap that is both positive and materially large (every detectable cell is either negative or below the 0.4% margin), so none would survive multiple-testing correction as a discovery.

Table 2: Spatial gap range (min, max over the nine cells, % of test CVaR_β) at three tail levels. The DRO is engaged ($\varepsilon^\star = 1$ almost everywhere) throughout. No level yields a sign-stable positive gap reaching the 0.4% materiality margin.

Grid	$\beta = 0.90$	$\beta = 0.95$	$\beta = 0.99$
Western US	[−0.04, +0.01]	[−0.01, +0.04]	[0.00, +0.14]
Eastern US–Canada	[−0.02, +0.04]	[−0.02, +0.04]	[−0.01, +0.09]
Diversified	[−0.23, +0.08]	[−0.23, +0.06]	[−0.36, +0.02]

The null is not a conditioning artifact. Block-diagonalizing Σ also changes its conditioning, so one might worry the null reflects the joint arm being harder to estimate rather than the cross-structure being valueless. The two arms are conditioned to the same order (ridge-regularized $\kappa \approx 1.1\text{–}1.6 \times 10^5$ for the joint covariance, $0.4\text{–}0.8 \times 10^5$ for the shuffled, the joint arm somewhat higher as expected for the richer model). The mean-ablation of Section 4.4 is the decisive control: it reuses the *same* joint Σ at the *same* conditioning and, removing only the mean field, the covariance recovers up to +1.46%. Estimation conditioning therefore cannot be what suppresses the spatial value; the mean field is.

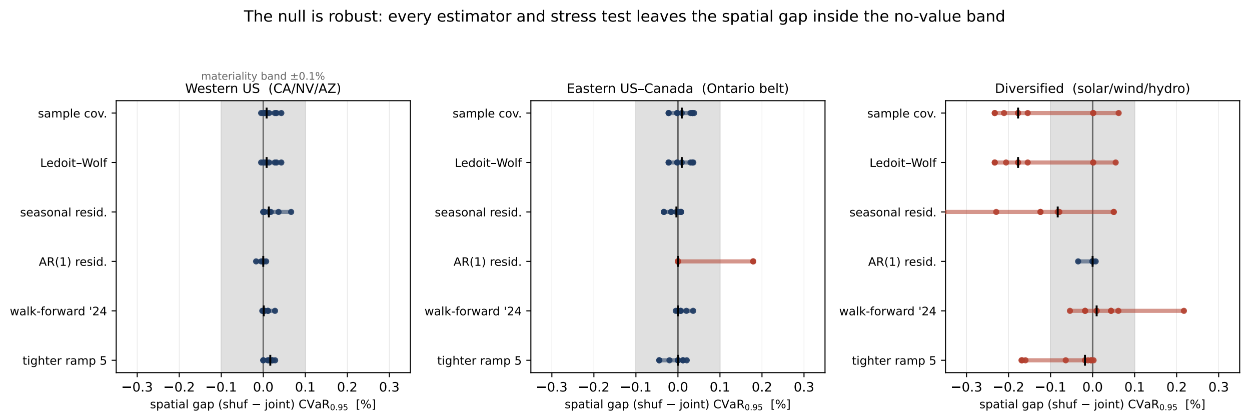

Figure 8: Robustness of the null. For each grid and each robustness arm (sample covariance, Ledoit–Wolf, seasonal and AR(1) residuals, walk-forward to 2024, and a 3× tighter ramp), dots are the spatial gap for the nine (regime, α) cells and the tick marks the median. The shaded band is the ±0.1% no-value threshold. No arm produces a sign-stable, material gap; the few Diversified excursions are negative (the joint covariance mildly hurts).

Statistical power: the null is not an underpowered test. A null is only as strong as the test's ability to reject it, so we quantify that ability directly. The paired day-bootstrap gives a standard error on the spatial gap of SE = 0.005–0.039% of $\mathrm{CVaR}_{0.95}$ across the three primary grids (Table 3). The minimum spatial benefit detectable at 80% power (two-sided, $\alpha = 0.05$) is MDE $= (z_{0.975} + z_{0.80})\,\mathrm{SE} = 2.80\,\mathrm{SE} = 0.015$–$0.11\%$. The test therefore had the power to detect a spatial advantage *an order of magnitude below* a ≈ 1% materiality threshold (the smallest gap a practitioner would act on; by the economic grounding of Section 5.4 this is ≈ 0.4% of the energy bill), and roughly the same order as the per-cell gaps themselves, yet no sign-stable positive gap of even this size survives. The null is a genuine absence of usable spatial value, not a test too blunt to see one.

Equivalence: the absence of a material effect is itself significant. Power bounds the effect the test *could* have missed; an equivalence test makes the complementary, affirmative claim that a material effect is *absent*. We apply the two-one-sided-tests (TOST) procedure [22, 23] *per cell*, using each cell's own paired-bootstrap standard error (recovered from its 95% gap interval), against the economic materiality margin $\Delta = 0.4\%$ of $\mathrm{CVaR}_{0.95}$ (Section 5.4; the ≈ 1% practitioner threshold is looser still), testing $H_0 : |\mathrm{gap}| \geq \Delta$ against $H_1 : |\mathrm{gap}| < \Delta$. All 27 cells across the three grids are equivalent: every 90% confidence interval lies strictly inside $(-\Delta, +\Delta)$. The binding cell is the adversarial Diversified case (gap = −0.233%, cell SE = 0.046%), which still clears both one-sided tests, $z = (\Delta - |\mathrm{gap}|)/\mathrm{SE} = 3.7$ ($p = 1.3 \times 10^{-4}$); every other cell gives $z \geq 7$. We therefore do not merely fail to detect a spatial benefit; we statistically reject the existence of one (in either direction) larger than the materiality threshold, cell by cell. That is the strongest form a null can take.

The claim is of course threshold-sensitive, so we state where it breaks. The smallest margin each grid still clears (its largest |gap| + 1.645 SE over cells) is $\Delta^\star = 0.05\%$ for Eastern US–Canada and 0.10% for Western US, but $\Delta^\star = 0.31\%$ for the adversarial Diversified grid. Equivalence at the 0.4% economic margin therefore holds everywhere with room to spare on the two interconnected grids, and only narrowly on the engineered low, heterogeneous-correlation portfolio. That last caveat does not threaten the thesis: the Diversified gap is *negative*, so the one grid where equivalence is hardest to assert is precisely the one where the joint covariance *hurts* rather than adds value.

Table 3: Statistical power of the spatial-gap test. SE is the paired-bootstrap standard error of the gap; $\mathrm{MDE}_{80} = 2.80\,\mathrm{SE}$ is the smallest true gap detectable at 80% power ($\alpha = 0.05$, two-sided). All are well below the $\approx 1\%$ materiality threshold.

Grid	SE (% $\mathrm{CVaR}_{0.95}$)	MDE_{80} (%)
Western US	0.036	0.10
Eastern US–Canada	0.005	0.015
Diversified	0.039	0.11

4.4 Mean-ablation: the covariance is masked, not worthless (RQ3)

The ablation isolates the cause (Table 5, Figure 6B). Under LEVEL ablation (regional time-averages equalized, hour-of-day shape kept) every gap is unchanged to three decimals: the *between-region* mean differences are not what masks the covariance. Under FLAT ablation (constant scheduling mean, the covariance-only world) the joint covariance suddenly pays: up to +1.46% in Diversified and +0.41% in Eastern US–Canada, all BH-significant, and CV no longer collapses both arms onto the same schedule. The spatial signal in the covariance is therefore *real and exploitable*; in the real problem it is simply dominated by the within-day shape of the mean carbon field, which both arms share. The instructive exception is Western US: even in the covariance-only world its gaps stay at zero (−0.036 to +0.004%), because its strong correlation is *common-mode*, all five zones move together, so destroying cross-dependence changes no scheduling decision. Positive correlation is non-diversifiable risk: there is no hedge for a DRO to find.

4.5 Tail dependence: the structure is non-elliptical (RQ3)

Table 6 reports residual tail-dependence at $q = 0.95$ for representative pairs. Two findings. *First*, the upper tail is empirically at or below the Gaussian benchmark everywhere (excess −0.18 to +0.04): in the dirtiest hours, regions *de-couple* rather than spike together, there is no upper-tail

Table 4: Robustness battery: maximum |gap| (%) per covariance estimator, and temporal walk-forward (test year 2024). No cell passes the pre-registered agreement rule (positive and sign-stable under both seasonal and AR(1) residualization).

Case	Ledoit–Wolf	Seasonal	AR(1)	Walk-forward 2024
Diversified	0.233	0.355	0.035	0.217
Eastern US–Canada	0.037	0.034	0.179	0.036
Western US	0.043	0.066	0.017	0.028

Table 5: Mean-ablation: spatial gap range (%) across the nine cells per case. LEVEL (equalize regional time-averages) leaves the null untouched; FLAT (constant scheduling mean = covariance-only world) reveals genuine spatial value except where correlation is common-mode (Western US).

Case	Real problem	LEVEL	FLAT (covariance-only)
Diversified	$-0.233\ldots+0.062$	unchanged	$+0.39\ldots+1.46$
Eastern US–Canada	$-0.022\ldots+0.038$	unchanged	$+0.14\ldots+0.41$
Western US	$-0.005\ldots+0.043$	unchanged	$-0.04\ldots+0.00$

co-movement for a richer ambiguity set to exploit on the risk side. *Second*, the dependence is radially asymmetric, $\chi_L > \chi_U$: regions go *clean* together more strongly than dirty together, clearest in the solar-driven US West (CISO–LDWP $\chi_L = 0.48$ vs. $\chi_U = 0.25$; BANC–LDWP 0.40 vs. 0.17). Elliptical copulas force $\chi_L = \chi_U$, so this is structure a Mahalanobis–Wasserstein ball is geometrically incapable of representing, the covariance is not merely masked (Section 4.4); for the part of the dependence that varies, it is also the *wrong object*. Figure 9 shows the asymmetry for the Ontario–Eastern belt.

4.6 Phase 2: the null survives richer dependence models (RQ3)

If the covariance ball failed only because it is elliptical, a copula that encodes the empirical $\chi_L > \chi_U$ asymmetry should recover value. It does not. Figure 10 confirms the Clayton copula fitted to each grid behaves as intended, its samples pile into the joint-clean corner ($\chi_L = 0.70$ vs. $\chi_U = 0.26$ for CISO–BANC), exactly the structure the symmetric Gaussian copula ($\chi_L \approx \chi_U \approx 0.46$) and any covariance ball cannot represent. Yet when each dependence model is used to schedule and evaluated out-of-sample on 2025 (Table 7, Figure 11), no fitted copula beats the independence baseline by a material, sign-stable margin; the Gaussian and Clayton gains never exceed 0.16% of $\mathrm{CVaR}_{0.95}$ and flip sign across α exactly as in Phase 1.

To test the *actual* worst case we add the comonotone (upper-Fréchet) arm: perfect rank coupling. Since CVaR is comonotone-additive and the upper Fréchet bound is the most positively-associated coupling consistent with the marginals, comonotone maximizes the CVaR of the resulting sum over

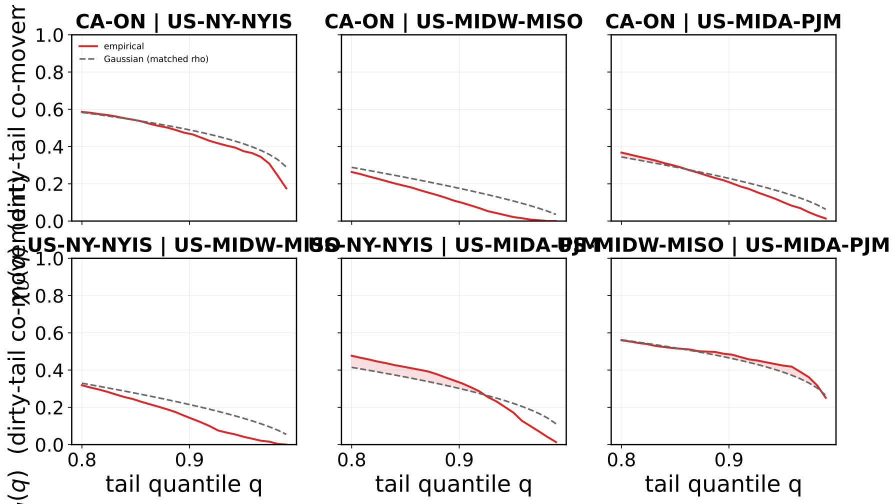

Figure 9: Upper-tail dependence $\chi_{\mathrm{U}}(q)$ for the Ontario–Eastern belt region pairs (solid) against the Gaussian-copula benchmark at the matched correlation (dashed). The empirical curves sit at or below the benchmark as $q \to 1$: in the dirtiest hours the regions decouple, so there is no exploitable upper-tail co-movement for a richer ambiguity set to capture. The complementary radial asymmetry ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$) is quantified in Table 6.

all copulas [4, 13], so it realizes the supremum $\sup_C$ in the leverage Λ of Section 3.9. This is the most any copula can buy. At the locked scenario seed its largest single-cell gain is +0.18% (Eastern US–Canada), and a naive day-bootstrap would call several Eastern US–Canada cells "detectable." That detectability is illusory, however: the bootstrap resamples test days but not the *scenario draw*, and re-running the comonotone arm across five scenario seeds shows the gap is sampling noise, ranging from −0.08% to +0.10% and centred on zero (mean +0.01%; Appendix A, Figure 13). The honest reading is therefore the stronger one: even the maximal copula's leverage Λ sits at the scenario-sampling *noise floor*, $|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$ and statistically indistinguishable from zero. This measurement is *independent* of the mean-ablation, so it pins Λ without circularity: no copula, not even the maximal one, recovers material value.

One positive signal is worth stating precisely: the Clayton−Gaussian increment is consistently positive in the heterogeneous case (Diversified, mean +0.07%) and negligible or negative where correlation is common-mode. The asymmetric copula therefore *does* extract something the elliptical one cannot, in the expected direction, but at a magnitude an order below materiality. This is the empirical face of Proposition 1: the copula's leverage Λ is real but dominated by the mean field. The clearest statement of the result is out of sample (Figure 12): the distributions of real 2025 daily emissions under the independence, Clayton, and comonotone schedules lie on top of one another, with $\mathrm{CVaR}_{0.95}$ values agreeing to the third significant figure. Phase 2 thus converts the Phase 1 null from "covariance is the wrong object" into the stronger "*no* dependence model recovers value here, because the binding constraint is the mean field, not the shape of the dependence."

Table 6: Residual tail dependence at $q = 0.95$ for selected pairs. "Excess" = empirical χ_U minus the Gaussian-copula value at the matched correlation ρ. Throughout, χ_U shows no exploitable upper-tail excess while $\chi_L > \chi_U$ signals non-elliptical (radially asymmetric) dependence.

Case	Pair	ρ	χ_U emp	χ_U Gauss	Excess	χ_L emp
Western US	CISO–LDWP	0.72	0.25	0.41	−0.16	0.48
Western US	BANC–LDWP	0.65	0.17	0.35	−0.18	0.40
Western US	CISO–NEVP	0.78	0.43	0.47	−0.05	0.39
Eastern US–Canada	CA-ON–NYISO	0.73	0.38	0.42	−0.04	0.31
Eastern US–Canada	MISO–PJM	0.70	0.43	0.39	+0.04	0.42
Diversified	CISO–BPAT	0.33	0.10	0.16	−0.06	0.26
Diversified	ERCOT–BPAT	0.00	0.04	0.05	−0.02	0.02

Table 7: Phase 2 copula results. Gap = reduction in out-of-sample $\mathrm{CVaR}_{0.95}$ versus the independence baseline (%; positive = the copula helps), the maximum *positive* gain over the nine cells at the locked scenario seed. The *comonotone* arm is the upper Fréchet bound (perfect rank coupling), the supremum over all copulas, hence the empirical ceiling Λ. The final column reports the comonotone gap re-evaluated over five scenario seeds: in every case it is centred on zero and sampling-noise dominated, so no gain is stable or material.

Case	$\bar{\tau}$	θ	$\text{gain}_{\text{Gauss}}$	$\text{gain}_{\text{Clay}}$	$\text{gain}_{\text{comon}}$	comon. over 5 seeds
Western US	0.47	1.77	0.066	0.038	0.089	≈ 0, noise
Eastern US–Canada	0.31	0.90	0.158	0.014	0.182	[−0.08, +0.10], mean +0.01
Diversified	0.20	0.50	0.069	0.127	0.114	≈ 0, noise

5 Discussion

5.1 What the null means, and what it does not

The result is a *specification* finding, not a failure of carbon-aware scheduling. Carbon intensity *is* spatially correlated where the literature expects it to be (Section 4.1), and the DRO machinery *is* active (CV selects $\varepsilon^\star = 1$ almost everywhere). What the experiment falsifies is the specific hypothesis that encoding that correlation in a covariance-based ambiguity set improves robust scheduling. The mean-ablation result (Section 4.4) is what licenses the stronger claim: the covariance signal is genuine and, in a covariance-only world, worth up to 1.46% of tail emissions; it is simply dominated by the diurnal mean carbon field in the problem as actually posed. A reader could otherwise dismiss the null as "the data had no signal"; the ablation shows the signal is present but *subordinate*. This is the central intellectual contribution: a clean separation of *validity* of a modelling assumption from its *value*.

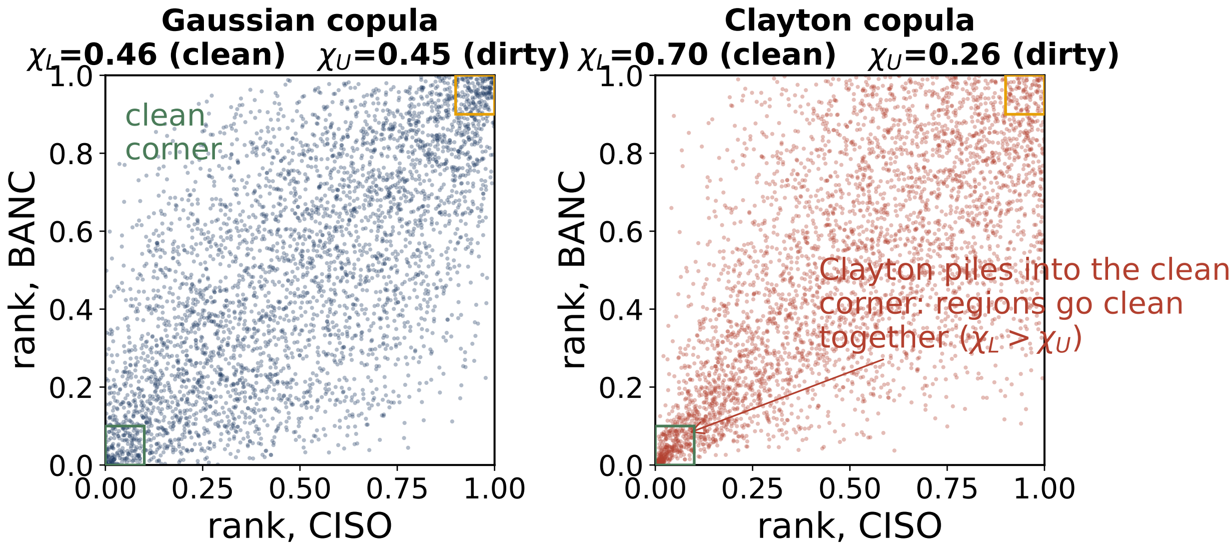

Figure 10: The dependence object a covariance ball cannot represent. Samples from the Gaussian copula (left, symmetric: $\chi_{\mathrm{L}} \approx \chi_{\mathrm{U}}$) and the fitted Clayton copula (right) for a representative US-West pair. Clayton concentrates in the lower-left "clean" corner ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), reproducing the Phase 1 radial asymmetry that elliptical models force to zero.

5.2 Why the mean dominates: a mechanism

Three forces compound. (i) *Magnitude.* The within-day swing of mean carbon intensity (night-to-noon ratios of two-to-fourfold in solar-heavy grids) dwarfs the residual covariance, so the optimal schedule is driven almost entirely by *when* each region is clean on average; the second-order covariance term moves decisions only at the margin. (ii) *Common-mode correlation.* Where correlation is strongest (US West), it is positive and shared across all zones: common, non-diversifiable risk. Diversification value requires *offsetting* fluctuations; positively co-moving regions offer no hedge, so destroying the cross-blocks (the shuffled arm) changes nothing. (iii) *Wrong tail.* Robust *tail* scheduling cares about joint *dirty* excursions, but the empirical upper tail is independent-to-decoupling (χ_{U} at or below Gaussian), while the co-movement that does exist is in the *clean* tail ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), exactly the regime a risk-averse scheduler is not protecting against, and a radial asymmetry an elliptical ball cannot encode regardless. The covariance is thus both *masked* (dominated by the mean) and, for its varying part, *mis-shaped* (non-elliptical). The same mechanism surfaces from the operational side: when these grids are scheduled under tightening geographic and ramp constraints, the binding dimension is geographic flexibility rather than temporal smoothing, again locating the value in *where* each region is clean on average rather than in second-order co-movement.

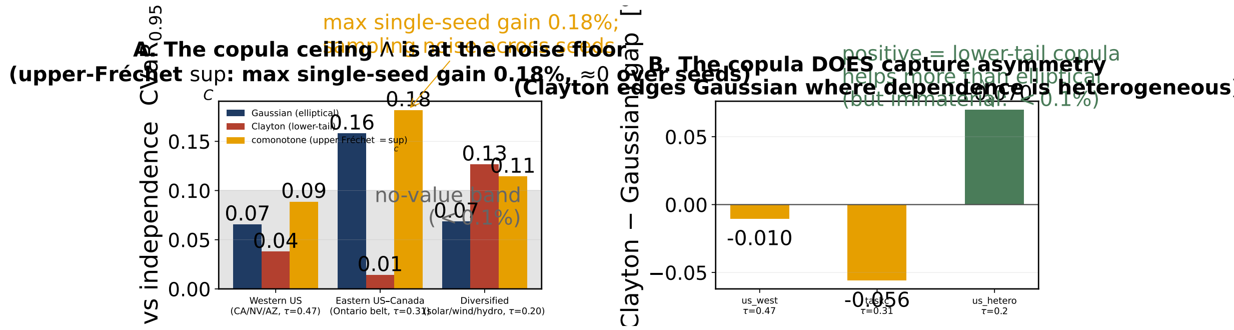

Figure 11: Phase 2 result (single locked scenario seed). **A:** maximum positive gain versus independence for the Gaussian, Clayton, and comonotone (upper-Fréchet $\sup_C$) arms. The largest single-seed comonotone gain is 0.18% (Eastern US–Canada); re-evaluated over scenario seeds it is sampling noise centred on zero (Figure 13). **B:** the mean Clayton−Gaussian increment by grid; positive where dependence is heterogeneous (Diversified, +0.07%), confirming the copula *does* capture the asymmetry, but an order of magnitude too small to matter, as Proposition 1 predicts.

5.3 Relation to prior work

The finding refines rather than contradicts the carbon-aware DRO literature. Single-region Wasserstein DRO for carbon-aware scheduling is established by Hall et al. [6], who report robustness gains over sample-average approximation; the present model is the natural *multi-region* extension that treats carbon intensity as a correlated vector. Two quantitative observations place our null in that context. First, the robustification *is* engaged: cross-validation selects a non-trivial ambiguity radius ($\varepsilon^\star = 1$, Figure 5) rather than collapsing to the nominal problem, consistent with the motivation in [6]. Second, holding that single-region robustification fixed, the cross-region covariance adds nothing: the spatial gap is null throughout. Indeed, in our out-of-sample 2025 test even the single-region DRO improves on the nominal (mean-greedy) schedule by at most 0.18% of $\mathrm{CVaR}_{0.95}$ across grids, itself a reflection of the mean-dominance we document; the spatial extension then adds a further nothing. The contribution is to map where the value is *not*: neither robustification depth nor spatial coupling materially improves on a mean-greedy per-region schedule for this problem. This saves practitioners from modelling complexity that does not pay and gives theorists a precise target: the value, if any, lives in non-elliptical tail structure that a Wasserstein ball over covariance cannot reach.

5.4 Practical recommendation

For carbon-aware schedulers on the grids studied, a *per-region marginal* robust scheduler, one ambiguity set per zone, no cross-region covariance, captures essentially all the robust value attainable *without an active transfer channel* (Appendix B), at a fraction of the modelling and estimation cost. Estimating a full joint covariance is not merely unnecessary; in the diversification-friendly case it

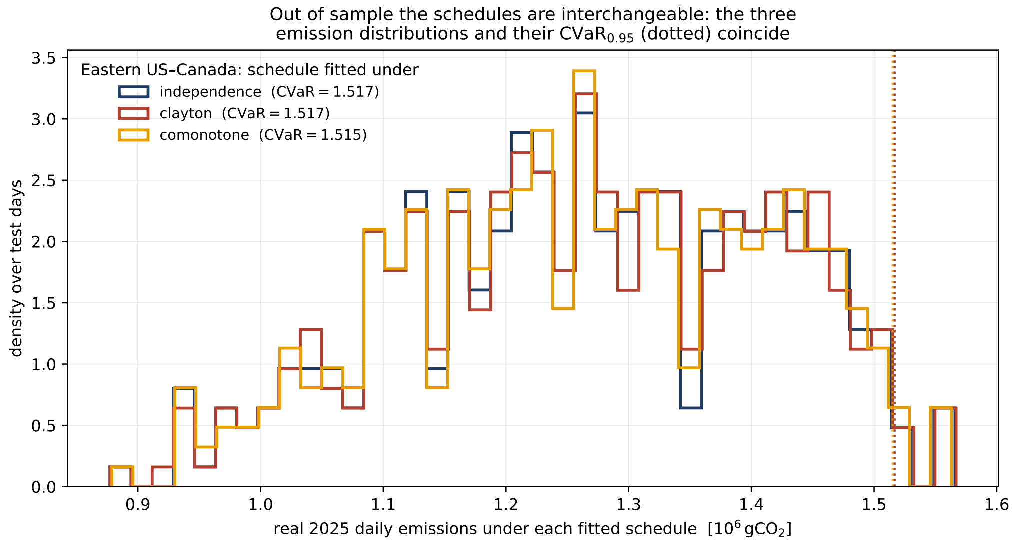

Figure 12: Out of sample the copula schedulers are interchangeable (Eastern US–Canada). Distribution of real 2025 daily emissions under the schedule fitted to independence, Clayton, and comonotone scenarios; the dotted lines mark each $CVaR_{0.95}$. All three coincide.

is mildly *counterproductive* (the joint arm fit noise and *lost* up to 0.23%). The engineering effort spared by not modelling spatial covariance is better spent on accurate per-region mean forecasts, which is where the value demonstrably lies.

To put the threshold in concrete terms, take the Eastern US–Canada grid: the modelled facility (four zones at 50 MW each, 80% utilization, $\approx$ 160 MW aggregate) emits on the order of 460,000 tCO_2 per year. A spatial-covariance benefit of 0.1% of $CVaR_{0.95}$, taken at its full point estimate, is about 460 tCO_2 per year, or roughly \$37,000 at a \$80/tCO_2 price. That is 0.04% of the facility's ≈\$84M annual energy bill, and it is not even reliably positive. The covariance channel is therefore immaterial in absolute as well as relative terms; the carbon savings worth pursuing live in the mean field, not its second moment.

5.5 Limitations

Five bound the claims. *(i) Representative schedule.* We evaluate one canonical workload (splittable load, ramp limits). A sharply different cost geometry could in principle re-weight the covariance term. The constraint-tightness sensitivities (Section 4.3) address the most plausible cases: a 3× tighter ramp and a 50–95% utilization sweep both leave the null intact. A qualitatively different objective (e.g. hard per-job deadlines) remains untested. *(ii) Test horizon.* The primary read is a single held-out year (2025); walk-forward to 2024 reproduces the null, but multi-year rolling evaluation would strengthen external validity. *(iii) Dependence model.* The DRO uses a covariance (second-moment) ambiguity set by construction; Phase 2 (Section 4.6) closes this gap with Gaussian and Clayton copula schedulers, but a full copula-*ambiguity* Wasserstein DRO in the Fan–Ji–Lejeune sense remains future work. *(iv) Geographic scope and the limit of generalization.* The region sets span WECC, the Eastern Interconnection, and Iberia–France across the dependence spectrum, but grids with different generation mixes (e.g. hydro-thermal systems with strong seasonal storage) remain untested. More fundamentally, the result generalizes only to grids where the diurnal mean field dominates residual cross-region covariance: mean-dominance is an empirical regularity of carbon intensity (Section 5.2), not a universal law, and a grid whose residual covariance rivalled its mean swing could in principle behave differently. *(v) Data licence and clocks.* Carbon intensity is Electricity Maps lifecycle estimates on a common local clock per region set; alternative intensity methodologies or clock conventions could shift point estimates, though not by the order of magnitude separating the observed gaps from materiality.

6 Conclusions

6.1 Answers to the research questions

RQ1, Is carbon intensity spatially correlated in coupled grids? Yes. After removing the shared diurnal cycle, residual cross-region correlations reach 0.78 in the US West and 0.73 in the Ontario-anchored Eastern belt and survive de-seasonalization (Section 4.1). The premise of spatial DRO is empirically sound; the engineered heterogeneous portfolio confirms the diagnostic resolves a genuinely near-zero *pair* (ERCOT–BPAT, 0.00) alongside moderate ones (mean $\approx$ 0.2), not a single washed-out value.

RQ2, Does encoding that correlation in a covariance-based ambiguity set improve robust scheduling? No. Across three primary region sets spanning the dependence spectrum (and two corroborating panels), the out-of-sample $\mathrm{CVaR}_{0.95}$ spatial gap never exceeds a few tenths of one percent and is not sign-stable; the null survives Ledoit–Wolf shrinkage, seasonal and AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward validation to 2024, a log-denser ε grid, and the full tail range from $\mathrm{CVaR}_{0.90}$ to $\mathrm{CVaR}_{0.99}$ (Sections 4.2–4.3), and a per-cell equivalence test rejects any effect above the materiality margin in all 27 cells. In the most diversification-friendly case the joint covariance is mildly counterproductive. Critically, this is an *active* null: cross-validation selects a non-trivial ambiguity radius ($\varepsilon^\star = 1$) in nearly every cell of the US/Canada cases, so the DRO is genuinely robustifying and the spatial covariance still adds nothing, a stronger result than a null in which the robustification had collapsed to the nominal problem.

RQ3, Why? Two complementary mechanisms. A mean-ablation experiment proves the covariance signal is real and worth up to 1.46% of tail emissions in a covariance-only world, but is dominated by the diurnal mean carbon field in the problem as posed (Section 4.4). A tail-dependence analysis shows the residual dependence is non-elliptical, upper-tail-independent and radially asymmetric ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), and therefore structurally invisible to a covariance ball even where it does vary (Section 4.5). The covariance is both *masked* and *mis-shaped*. Phase 2 settles which mechanism binds: replacing the covariance ball with a Clayton copula built for the $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ asymmetry still recovers no material value (Section 4.6), and Proposition 1 explains why, CVaR's translation invariance makes the dependence model second-order to the mean field. The limit is not the shape of the dependence object; it is mean-dominance.

6.2 Contributions

This thesis contributes: (1) a pre-registered falsification methodology (shuffled-marginals, extended to copula-coupled resampling) that cleanly separates the *validity* of a spatial modelling assumption from its *value*, transferable to any correlated-uncertainty DRO; (2) a replicated, robustness-checked null on whether spatial dependence improves carbon-aware DRO scheduling, holding across multiple real grids *and* across the dependence hierarchy from covariance ball to elliptical and lower-tail copulas; (3) a causal mechanism (formalized as a mean-dominance bound (Proposition 1) and demonstrated by mean-ablation) that explains the null rather than merely reporting it; and (4) a fully reproducible, version-controlled, CI-tested pipeline (194 tests) with archived license-safe summary tables for every reported number.

6.3 Future work

Phase 2 closed the most immediate question, whether a copula that sees the asymmetry recovers value (it does not), and in doing so sharpened the remaining agenda. First, a *copula-ambiguity Wasserstein DRO* in the Fan–Ji–Lejeune sense [3], built on a fitted vine [9, 10], which robustifies the dependence structure itself rather than sampling from a single fitted copula; Proposition 1 predicts the mean field will still dominate, but this would make the claim fully model-free. Second, and most promising, a *spatially-coupled transfer DRO*: introduce inter-region load flows $f_{r\to s,t} \geq 0$ as decision variables, so that the work served in a zone, $\tilde{x}_{r,t} = x_{r,t} + \sum_s (f_{s\to r,t} - f_{r\to s,t})$, can migrate toward whichever region is cleanest at each hour, subject to flow conservation, transfer-capacity and (optionally) transmission-loss constraints. Minimizing the worst-case CVaR of $\langle \rho, \tilde{x} \rangle$ over a Wasserstein ball keeps the program an SOCP, but now the spatial structure enters the *decision*, not merely a passive penalty. The motivation is precise: the mean field dominates (Proposition 1), and it dominates *spatially* too, since at each hour one region is cleanest on average. A passive ambiguity set cannot act on that first-order spatial signal; an active transfer channel can, which is exactly the value our null leaves on the table. The honest open question, and the real content of the next phase, is not whether transfer helps (it must, being mean-exploiting) but whether *robustifying* it beats plain deterministic transfer, e.g. under forecast error or for tail-averse operators, since the same mean-dominance logic suggests the robustness layer may again be second-order. Preliminary experiments on a unit-tested transfer module (Appendix B) already bear this out and sharpen it: active transfer cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.7–10.1%; robustness stays immaterial under normal carbon; but a stylised grid-emergency stress reveals a sharp tail-risk *crossover* beyond which the robust commitment wins, by up to +8%. To our knowledge no prior work combines Wasserstein-DRO robustification with active inter-region transfer; the channel overlaps a collaborator's deterministic-transfer scope and awaits supervisor sign-off. Further out, the transfer model invites an *online, multistage* treatment in which forecasts arrive sequentially and demand and transmission are themselves uncertain, with a Wasserstein ball centred on a learned forecast, and the mean-dominance bound invites generalization into a problem-class condition for *when* cross-region dependence can ever improve a robust allocation, of which carbon-aware scheduling is one instance. The null is thus not a dead end but a sharpened mandate: the value, if any, will not come from a better passive dependence model but from an active spatial decision.

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A Reproducibility Details

Scenario-count stability and the comonotone noise floor. Figure 13 re-evaluates the comonotone (upper-Fréchet) gap on Eastern US–Canada over five scenario seeds and a range of scenario counts S. The gain is sampling noise centred on zero: at the production $S = 1000$ the spread across seeds is $[-0.08, +0.10]\%$ with mean $+0.01\%$, and it narrows with S. This is the basis for treating the single-seed $+0.18\%$ of Section 4.6 as immaterial rather than a real effect, and it confirms $S = 1000$ is sufficient.

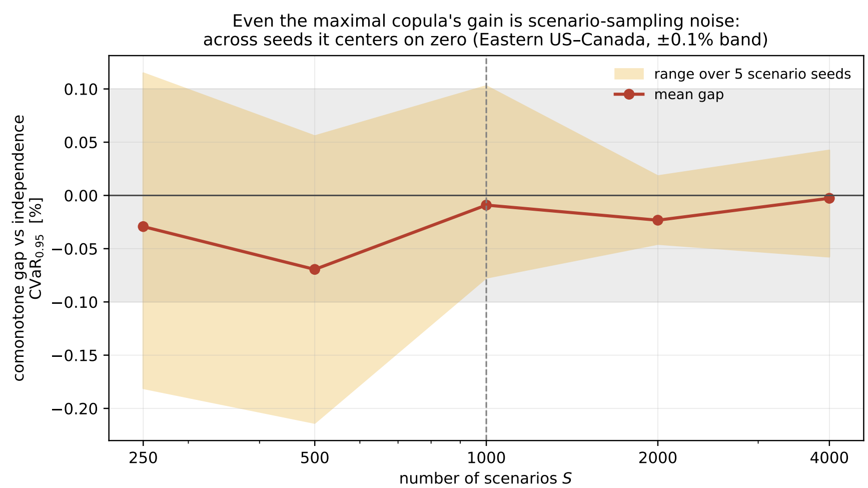

Figure 13: The comonotone gap is scenario-sampling noise. Mean (line) and range over five seeds (band) of the comonotone-vs-independence $\mathrm{CVaR}_{0.95}$ gap on Eastern US–Canada, against the scenario count S. The band straddles zero and narrows with S; the $\pm 0.1\%$ no-value band is shaded.

Repository and environment. All code lives in a private, version-controlled repository with a continuous-integration suite (175 unit tests; the Electricity-Maps integration test is excluded from CI as it requires licensed data). Experiments run under Python with CVXPY; solvers are CLARABEL/ECOS for the second-order cone program and HiGHS for deterministic LP baselines. The licensed raw carbon-intensity data is never committed; only derived aggregate summary tables (CV-selected ε, test CVaR, spatial gaps, bootstrap CIs, BH p-values) are archived, so the non-redistribution academic licence is respected while every reported number remains traceable.

Pre-registration protocol. Each experiment follows commit-lock → dry-run → single test read. The full configuration (utilization 0.80; $\alpha \in \{0.30, 0.50, 0.75\}$; ε grid $\{0, 0.1, 1, 10, 100, 1000\}$; 5-fold blocked time-series CV; 1000 bootstrap resamples at a fixed seed; CVaR level 0.95; train 2021–2024, test 2025) is committed before any test-set evaluation, and a -dry-run gate runs cross-validation only. The 3c capacity bounds $(x_{\min}, x_{\max}) = (50, 75)$ were Goldilocks-calibrated on training data alone (bind fraction 0.28–0.37).

Reproducing the experiments. Every result regenerates with

```
python -m scripts.run_case_experiment -region-set <case> [flags]
```

where <case> is one of the internal region-set keys us_west (Western US), taskc (Eastern US–Canada), or us_hetero (Diversified), and flags select the estimator (-shrinkage, -residualize), mean-ablation (-ablate-mean), walk-forward (-test-year), or grid density (-eps-grid). Snapshot files are named <case>_regimes_<date>[_<variant>].csv, and the snapshot README maps each to its source command. Table 8 maps each results subsection to its archived table.

Table 8: Mapping of reported results to archived summary tables.

Section	Result	Archived source
4.2	Spatial gaps (Table 1)	{case}_regimes_2026-06-10.csv
4.3	Robustness (Table 4)	{case}_..._{lw,seasonal,ar1,ty2024}.csv
4.3	BH correction	bh_correction.csv
4.4	Mean-ablation (Table 5)	{case}_..._ablate-{level,flat}.csv
4.1,4.5	CA/Iberia panels	taskA_.. es_pt_fr_..csv

B Preliminary Phase 3 results: active transfer and the tail-risk crossover

The work in this appendix lies beyond the scope of the capstone proper (Sections 1–6); it is a preliminary glimpse of a planned multi-part extension and is included only to show where the line leads. The capstone's graded contribution is the rigorous, pre-registered null and the mean-dominance bound; what follows is deliberately exploratory.

This appendix sketches where the line goes next, with *preliminary, proof-of-concept* experiments. They are prototype-grade, a stylised grid-emergency stress model, modest scenario counts, no pre-registration, and a rigorous treatment is the subject of a follow-on study; we include them only because they already complete the arc the capstone opened. The capstone showed that *passive* spatial modelling adds no value because the mean dominates. The natural question is whether making the spatial structure an *active decision* changes that. We add inter-region load flows $f_{r \to s,t} \geq 0$ so the executed load $y_{r,t} = x_{r,t} + \sum_s (f_{s \to r,t} - f_{r \to s,t})$ can migrate toward the cleaner region; the program stays an SOCP/LP.

Three findings emerge (Figure 14). **(1) Active transfer unlocks spatial value.** A transfer-budget sweep cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.7–10.1% across the three grids, the value the passive null left on the table, captured by exploiting the spatial *mean* (shipping work to the region cleanest on average). **(2) Robustness is worthless under normal carbon, even here.** A one-shot transfer DRO, a forecast-error (persistence) ambiguity set, and a two-stage robust commitment with costly recourse all leave robust ≈ deterministic, the mean-dominance of the main text extends into the active setting. **(3) But robustness pays under tail risk.** Modelling rare grid-emergency events (with probability 0.1 per day a region's carbon is scaled by a severity M, as in a renewable drought or peaker dispatch) and re-solving the two-stage problem with migration-limited recourse traces a sharp *crossover*: at $M = 1$ the robust commitment is no better than the risk-neutral one (the main null), but as M grows it increasingly wins, by up to +8.4% of $\mathrm{CVaR}_{0.95}$ at $M = 4$ on the US West. The robust commitment stays hedged while the risk-neutral one is burned when its favoured region spikes.

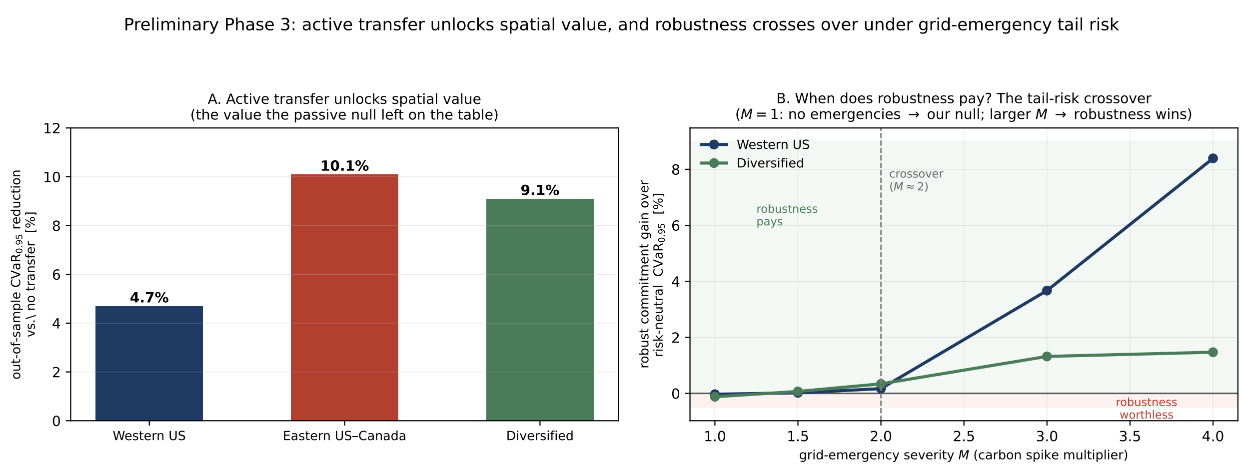

Figure 14: Preliminary Phase 3. **A:** active transfer reduces out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.7–10.1%. **B:** the tail-risk crossover, the robust (CVaR-hedged) commitment's advantage over the risk-neutral one as grid-emergency severity M grows; near $M \approx 2$ it turns positive. Proof-of-concept; full treatment is future work.

The synthesis is a clean phase diagram: distributional robustness is immaterial for carbon-aware scheduling under normal conditions (the mean dominates, whether the spatial structure is modelled passively or acted on actively), but becomes valuable precisely when rare tail events make the mean forecast unreliable. The value of robustness is thus a function of tail risk, the precise characterisation of which, together with a calibrated emergency model and a pre-registered evaluation, is the agenda for the follow-on work.

Implementation and validation. These results are not throwaway script output: the transfer model is implemented as a standalone, unit-tested module (`src/models/transfer_dro.py`), where the one-shot transfer DRO, the two-stage robust commitment, and the costly-recourse evaluation are each exercised by tests pinning the model's invariants, conservation of work under inter-region flows ($\sum_{r,t} y_{r,t} = \sum_r W_r$), monotone emission reduction as the transfer budget grows, and exact recovery of the no-transfer schedule at zero budget. The flow variables $f_{r\to s,t}$ carry no self-loops and respect a transfer-capacity budget, and the two-stage program is solved as an LP per scenario with a shared CVaR epigraph. What remains *preliminary* is therefore the experimental protocol, the stylised emergency model, the modest scenario counts, and the absence of pre-registration, not the optimization itself: the crossover of Figure 14 rests on validated code. This is what makes promoting Part 3 to a rigorous study a matter of protocol and calibration, the steps below, rather than of rebuilding the machinery.

The extension beyond this capstone. These preliminary results are the opening of a planned five-part programme that continues well past the capstone. The capstone is Parts 1–2 (the covariance and copula nulls plus the mean-dominance bound). **Part 3**, sketched above, is the spatially-coupled transfer DRO, the active-decision algorithm and the tail-risk crossover, promoted to a rigorous, pre-registered study with calibrated emergencies and, ideally, affine decision rules for the recourse policy. **Part 4** takes the transfer model *online*: a multistage / rolling-horizon robust controller in which forecasts arrive sequentially and demand and transmission are themselves uncertain, with a Wasserstein ball centred on a *learned* forecast (learning-augmented DRO) and evaluated closed-loop on real operational traces, the deployable systems contribution. **Part 5** generalises the mean-dominance bound itself into a problem-class condition for *when* cross-coordinate dependence can ever improve a robust allocation, of which carbon-aware scheduling is one instance and energy storage, EV fleets and spatial supply chains are others, the theory that turns the sequence into a research programme. The programme is detailed in the project's research roadmap; this appendix is only its first, deliberately preliminary, datapoint.

Annex A: Individual Contribution Statement

This Research Capstone was completed individually by **Marco Ortiz Togashi** under the supervision of Prof. Bissan Ghaddar. As sole author, I am responsible for all components of the work:

- **Research design.** Formulated the three research questions, designed the shuffled-marginals falsification test, and specified the pre-registration protocol that separates assumption validity from value.
- **Modelling.** Specified the Mahalanobis–Wasserstein DRO scheduling model and its SOCP reformulation, the constraint regimes (including the variable carbon-coupled capacity constraint adapted from the SoCC'25 formulation), and the mean-ablation and tail-dependence diagnostics.
- **Data and implementation.** Assembled the multi-grid carbon-intensity and temperature panels, implemented the full experimental pipeline, the robustness battery, the Benjamini–Hochberg correction, and the figures, under version control with a continuous-integration test suite.
- **Analysis and writing.** Produced and interpreted all results, established the causal mechanism, and wrote the manuscript in full.

Generative AI tools were used as a coding and drafting assistant under my direction, as declared on the cover page; all research questions, modelling decisions, experimental designs, interpretations, and final text are my own. The inter-region transfer-channel proposal noted as future work overlaps a collaborator's scope and is identified as such, pending supervisor coordination.

Marco Ortiz Togashi	Date